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Session Objectives: At the conclusion of this lecture the student will be able to:

1. Distinguish the pathophysiology and treatment of the following cold injuries: trench foot; chilblains (pernio); panniculitis; cold urticaria; frostbite
2. Assess and manage patients in various stages of accidental hypothermia
3. Identify the poor prognostic signs which justify termination of resuscitative efforts in accidental hypothermia

Mainstays of Frostbite Treatment

- **Rapid** rewarming ONLY after re-freezing risk has been eliminated
- **Immersion in, or application of warm water at 37°C to 39°C** (98.6°F to 102.2°F) until affected area is pliable and erythematous;
- Opioid analgesia
- Tetanus immunization status should be assessed and appropriate vaccination administered if needed, because frostbite is a tetanus-prone wound

Frostbite Treatment

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- **Noncontroversial**
  - **Ruptured clear blisters should be debrided** - removes destructive thromboxane and prostaglandins; allows application of topicals.
  - **Hemorrhagic – leave these alone** → worse outcome if debrided (except if very large and over a joint). One study showed you could aspirate*
  - Intravenous/intra-arterial anticoagulation and **thrombolysis for severe frostbite** (1 C recommendation = strong recommendation with weak evidence)**

- **Controversial**
  - Unruptured clear blister debridement – theoretically, blister contents are sterile
  - Systemic antibiotics
  - Topicals - bacitracin; silver sulfadiazine; aloe vera
  - Methylprednisolone
  - Sympathetic blockade –
    - Surgical: sympathectomy
    - Medical: bupivacaine - a long-acting local anesthetic
  - Hyperbarics, heparin and dextran, – all unproven

*Frostbite: pathogenesis and treatment. AUMurphy JV, Banwell PE, Roberts AH, McGrouther DA SOJ Trauma. 2000;48(1):171.

**UpToDate, Frostbite: Acute Care and Prevention, Zafren K. et al, September 2025

Causes of Secondary Hypothermia (It's not what this lecture is about, but these exacerbate the problem)

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- **Increased Heat Loss**
  - Burns
  - Iatrogenic (i.e., blood transfusions and other cold infusions, cooling blankets, inadequate insulation)
  - Recent birth
- **Impaired Thermogenesis**
  - Impaired shivering (i.e., advanced or very young age, malnutrition, physical exhaustion, neuromuscular disease)
- **Multifactorial**
  - Medications and other xenobiotics i.e., alcohol (COMMON), anesthetic agents, opioids, sedatives, vasodilators)
  - Sepsis especially in elderly or cachectic patients (one of the sepsis criteria is temp. <36°C)
  - Metabolic and endocrine disorders (i.e., alcoholic or diabetic ketoacidosis, adrenal failure, hypoglycemia, hypopituitarism, hypothyroid, lactic acidosis, Wernicke's encephalopathy). These are RARE causes of hypothermia.
  - Neurologic
    - lesions affecting hypothalamus
    - spinal cord injury causing autonomic dysreflexia)
  - Shock
  - Trauma with blood loss

Ability to shiver is crucial in raising body temperature

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- Shivering can be suppressed by:
  - Medications (e.g., analgesics, sedatives)
  - Exhaustion of energy stores
  - The application of warming devices (e.g., hot packs; forced-air blankets; warm, humidified air)
  - A core temperature drop below a critical level (~31°C).
    - The suppression of shivering by various warming methods is one of the reasons why the rewarming literature is so controversial.

Management – Stage 1: 35-32°C (conscious and shivering)

- Warm environment and dry clothing
- No hot baths - can cause vasodilation and convective cooling
- Warm, sweet drinks (thermogenesis requires immediate calories)
- Active movement (if possible)
- Thermistor core temperature monitoring not needed

Management – Stage 2: ‹32-28°C (**impaired consciousness**, +/- shivering)

- Active external (warm environment; chemical, electrical, or forced-air heating; warm packs or warm blankets)
- Minimally invasive rewarming techniques
  - Warm (38-42°C) **parenteral** fluids;

  (airway is not likely to be protected with impaired consciousness, nothing by mouth (NPO-nil per os); warm, humidified oxygen; bladder or nasogastric irrigation with warm fluids.
- Thermistor core temperature monitoring
- Minimal and cautious movements to avoid arrhythmias
- Full-body insulation in the horizontal position
- Consider other causes of impaired consciousness – ETOH, head injury, infection etc.

Management – Stage 4: May occur at <32° but usually < 28°C (no BP or pulse present, no forward blood flow on ultrasound)

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- **CPR**
- Up to 3 doses of epinephrine and **defibrillation (if in ventricular fibrillation)**. Further dosing is guided by clinical response.
- Airway management
- Transport to ECMO/CPB.
- Only if ECMO is unavailable should pleural or peritoneal warm saline lavage be considered
- Active external WITH minimally invasive rewarming during transport is recommended
    - External rewarming by itself or rewarming the extremities without simultaneously rewarming the trunk MAY cause afterdrop
- do not apply heat to head

Exercise particular care with:

- Sedatives and analgesics – may cause vasodilation, but risk of benefit may outweigh risk of harm (yes, that's the way it should be phrased)

- Vasopressors – patients are likely maximally vasoconstricted early. Their use in later stages is controversial.
  - **Ones with beta1 effects (e.g dobutamine) are preferred over ones with alpha (e.g. norephineprine)**
- Central venous access – keep catheter away from irritable right ventricle.

Afterdrop

- As periphery warms and vasodilates, warmer core blood is shunted to the periphery
- Studies show it’s minimal*

*Giesbrecht GG, Goheen MS, Johnston CE, Kenny GP, Bristow GK, Hayward JS: Inhibition of shivering increases core temperature afterdrop and attenuates rewarming in hypothermic humans. *J Appl Physiol (1985)* 83: 1630, 1997.

*Giesbrecht GG, Bristow GK: The convective afterdrop component during hypothermic exercise decreases with delayed exercise onset. *Aviat Space Environ Med* 69: 17, 1998.

These are very poor prognostic signs which support the decision to terminate resuscitative efforts:

- Serum **K+ > 12** mmol/L
- **Rewarmed >32°C** and still with no forward blood flow
- Ultrasound shows **no forward flow of blood**? **Vasopressors will not likely have any effect.**
- Obvious mortal injury e.g. decapitation, decomposition
- Witnessed prior normothermic arrest
- Drowning is a bad prognosis. Case reports of survival usually involve very young children with intact diving reflexes.
- Frozen solid
- If none of the above, then RECONSIDER an in place DNR order, because in early stages of hypothermia the prognosis is better